Predicting Star Types

Bradley, Siera, Haydn, Mamadou

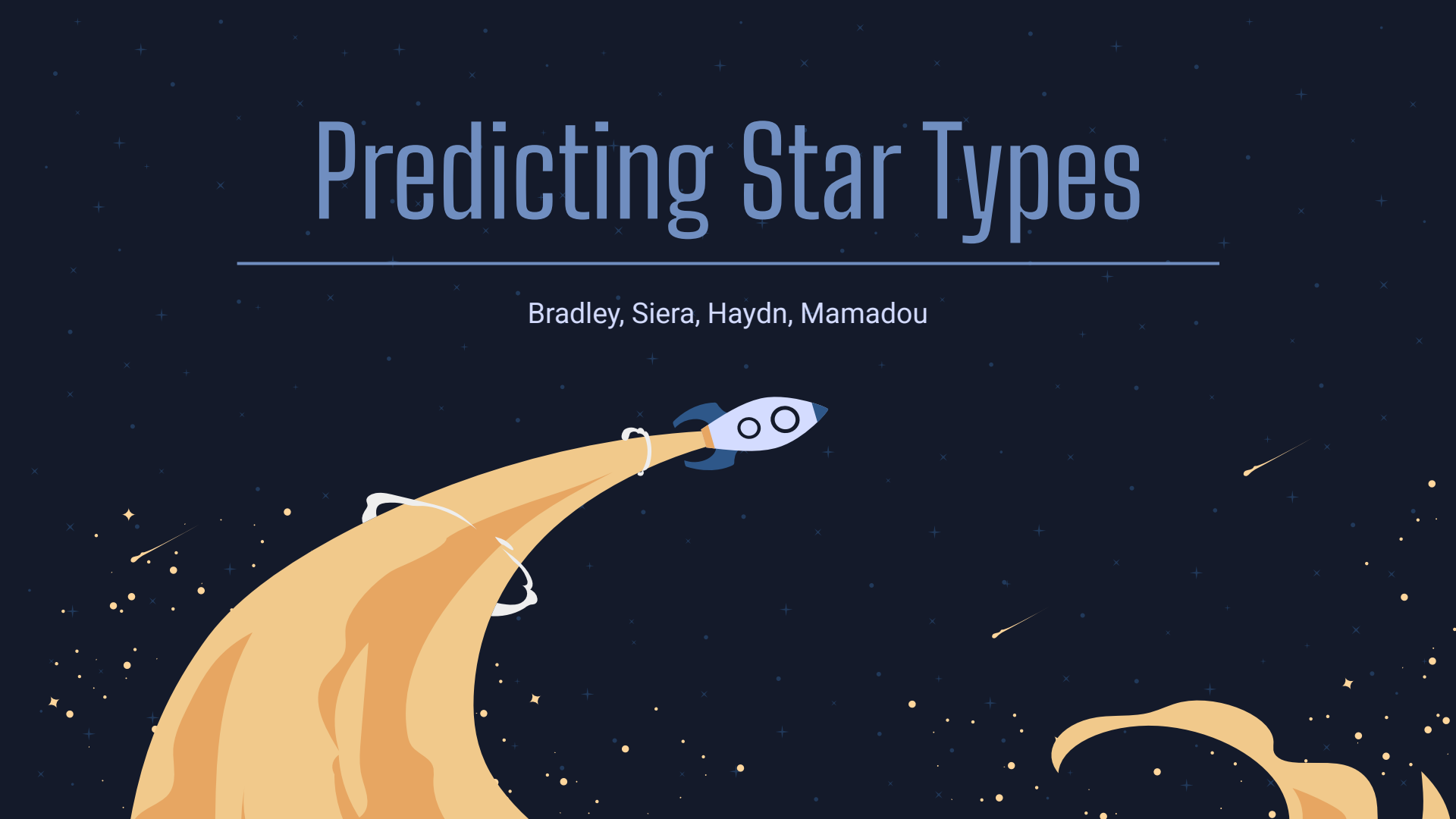


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01

Our Data

Source and Wrangling

Types and magnitude

Source

Kaggle
“Star Dataset”
240 rows, 7 columns

Significant Variables

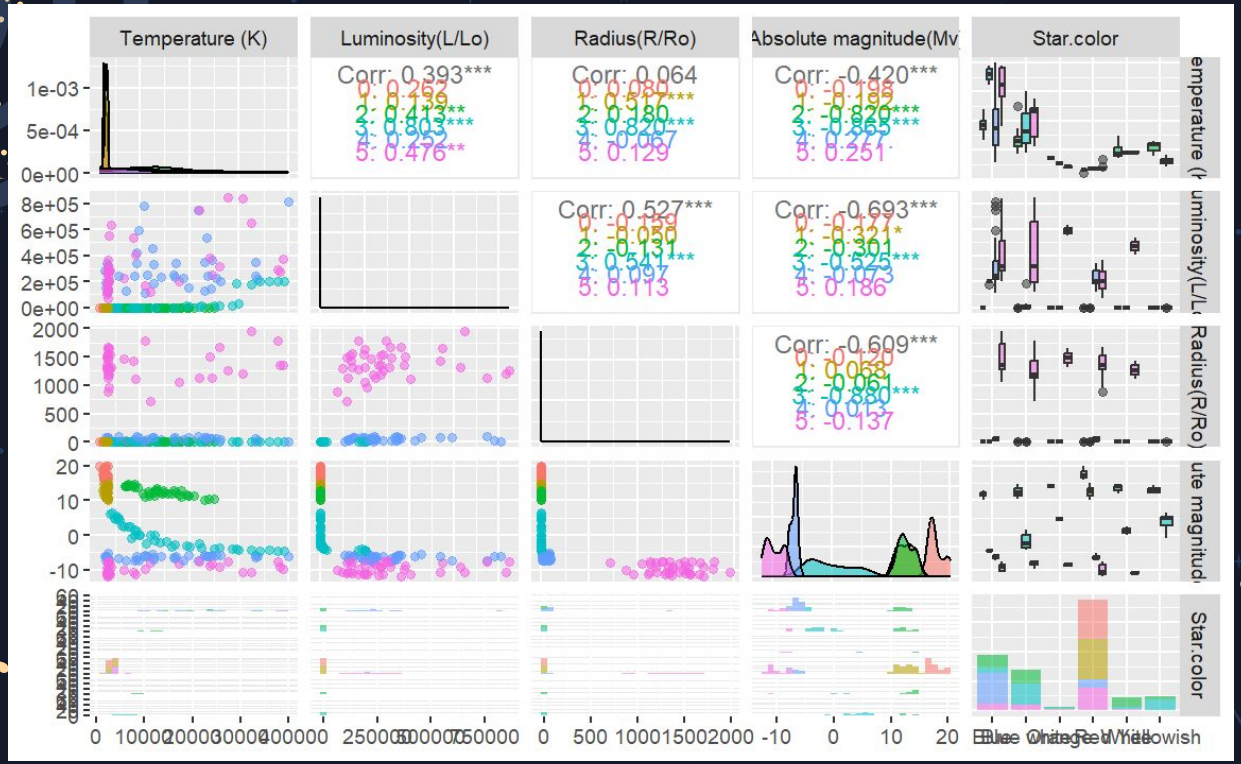
Temperature (K)
Luminosity (L/Lo)
Radius (R/Ro)
Absolute Magnitude
Star Color
Star Type

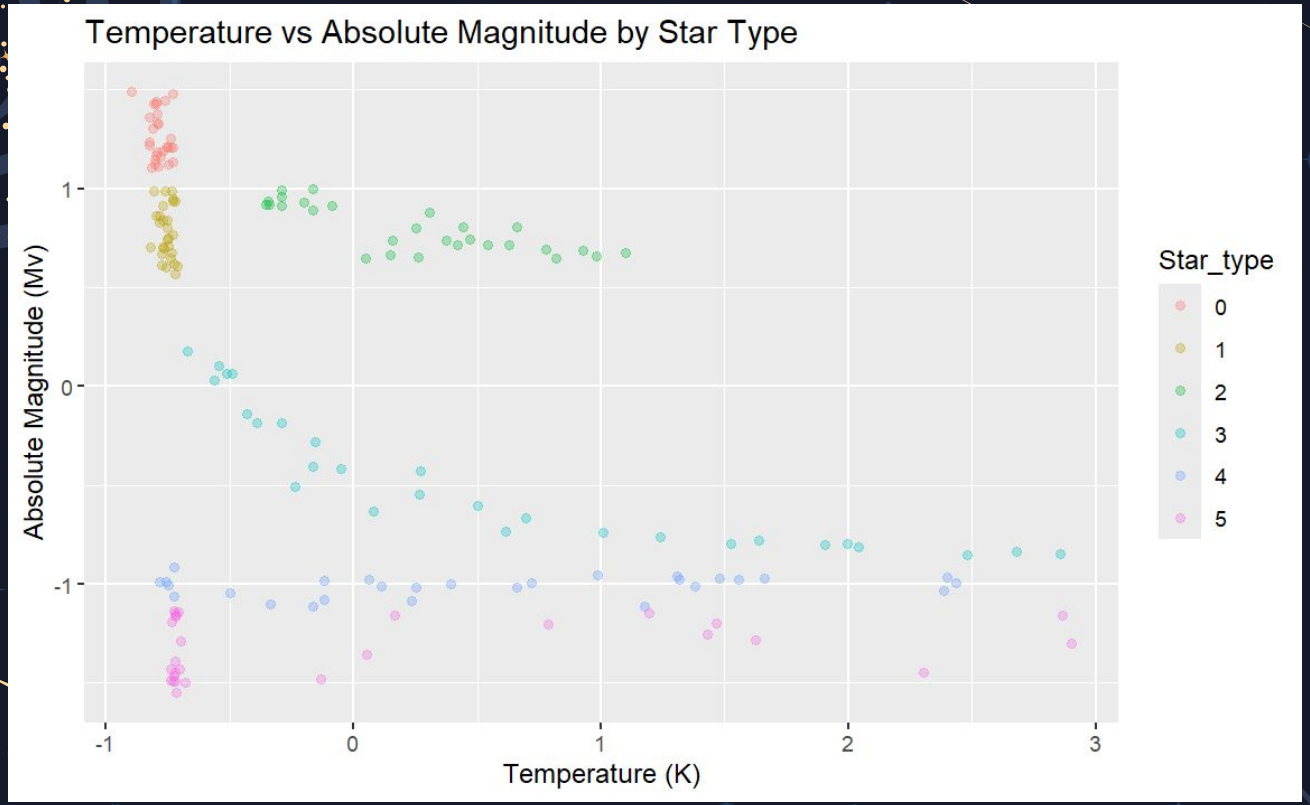
Wrangling

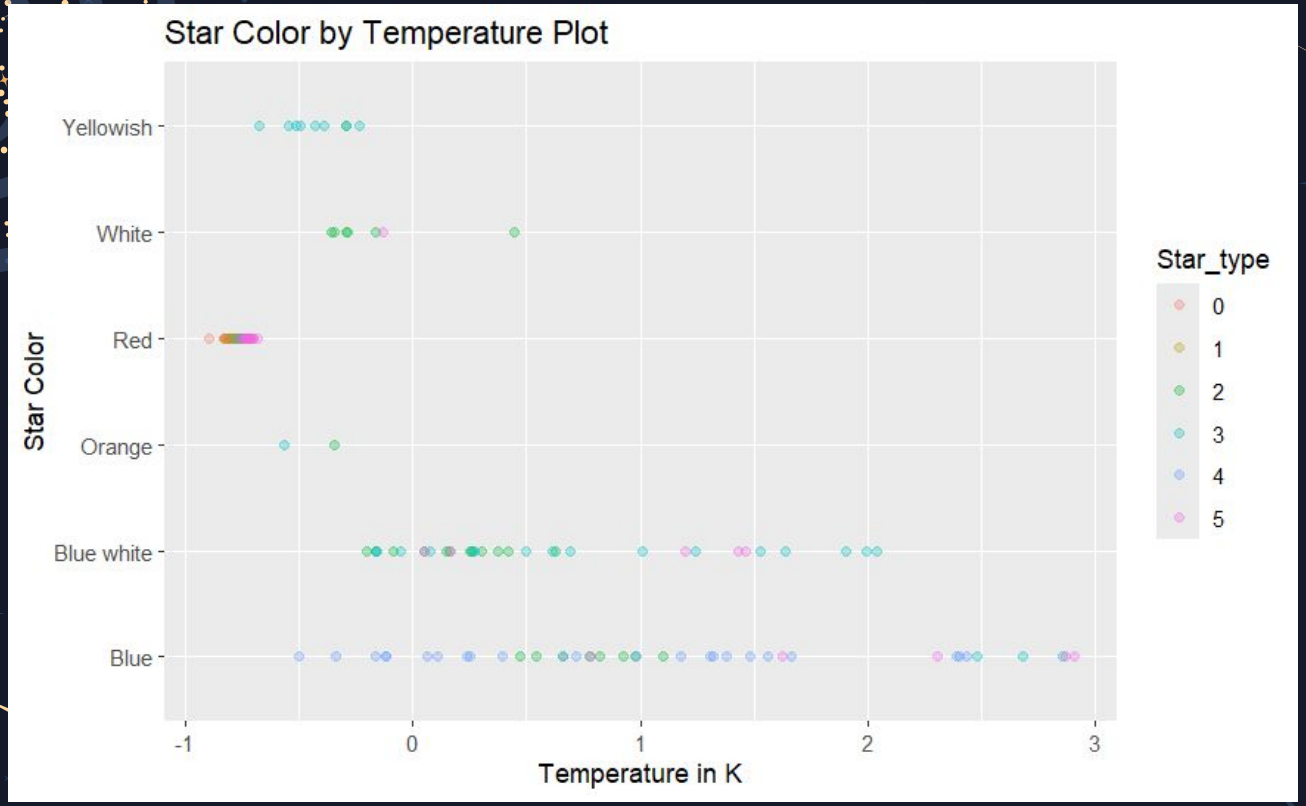
Rename all color types.

Remove spectral class variable

Create test + train datasets (30-70)


Temperature (K)
Luminosity(L/Lo)
Radius(R/Ro)
Absolute magnitude(Mv
Star.color
Corr: 0.393***
Corr: 0.064
Corr: -0.420***
Corr: 0.527***
Corr: -0.693***
Corr: -0.609***
1e-03
5e-04
0e+00
8e+05
6e+05
4e+05
2e+05
0e+00
2000
1500
1000
500
0
20
10
0
-10
0
10000
20000
30000
40000
250000
500000
750000
500
1000
1500
2000
-10
0
10
20
Blue
Orange
Red
White
Yellowish


Temperature vs Absolute Magnitude by Star Type
Absolute Magnitude (Mv)
1
0
-1
-1
0
1
2
3
Temperature (K)
Star_type
0
1
2
3
4
5

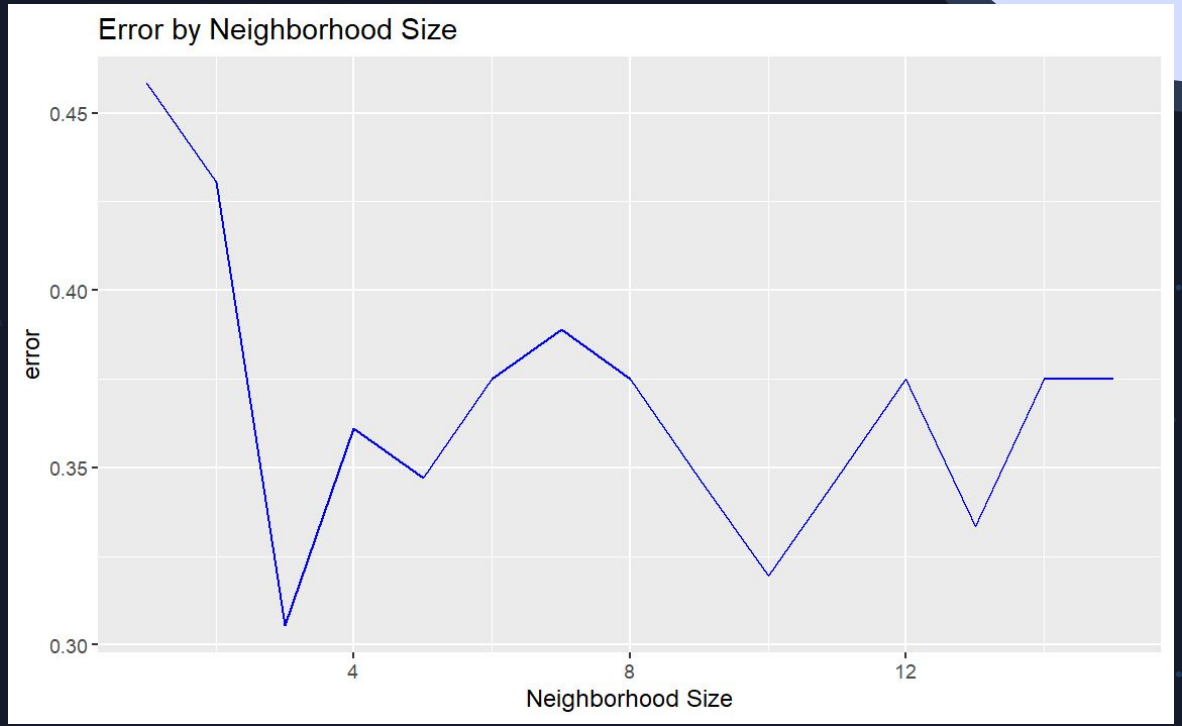

Star Color by Temperature Plot
Yellowish
White
Red
Orange
Blue white
Blue
Star Color
-1
0
1
2
3
Temperature in K
Star_type
0
1
2
3
4
5

02

KNN Model



Picking Neighborhood Size



KNN Prediction

Accuracy: 69.4%

Using:

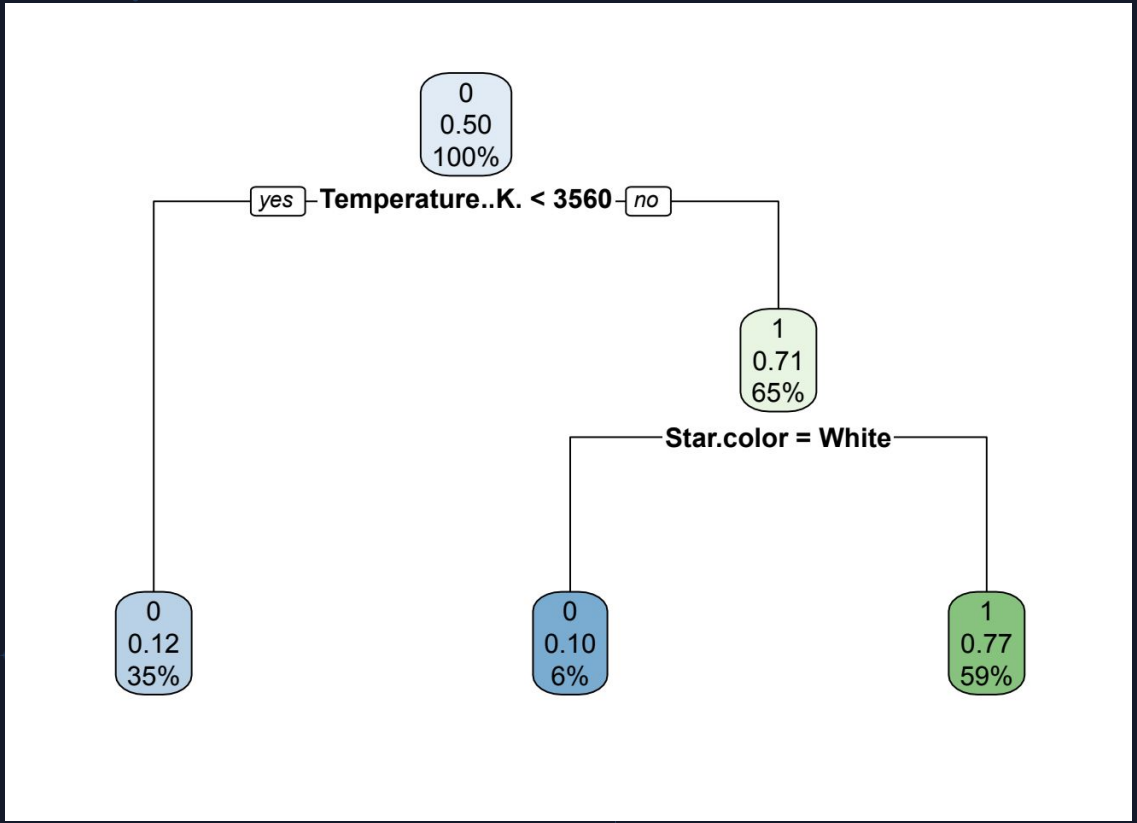
- Temperature
- Luminosity
- Star Color

stars_pred	0	1	2	3	4	5
0	8	2	0	0	0	0
1	4	10	0	0	0	0
2	0	0	7	4	0	0
3	0	0	5	8	0	0
4	0	0	0	0	7	2
5	0	0	0	0	5	10

03

Tree Model

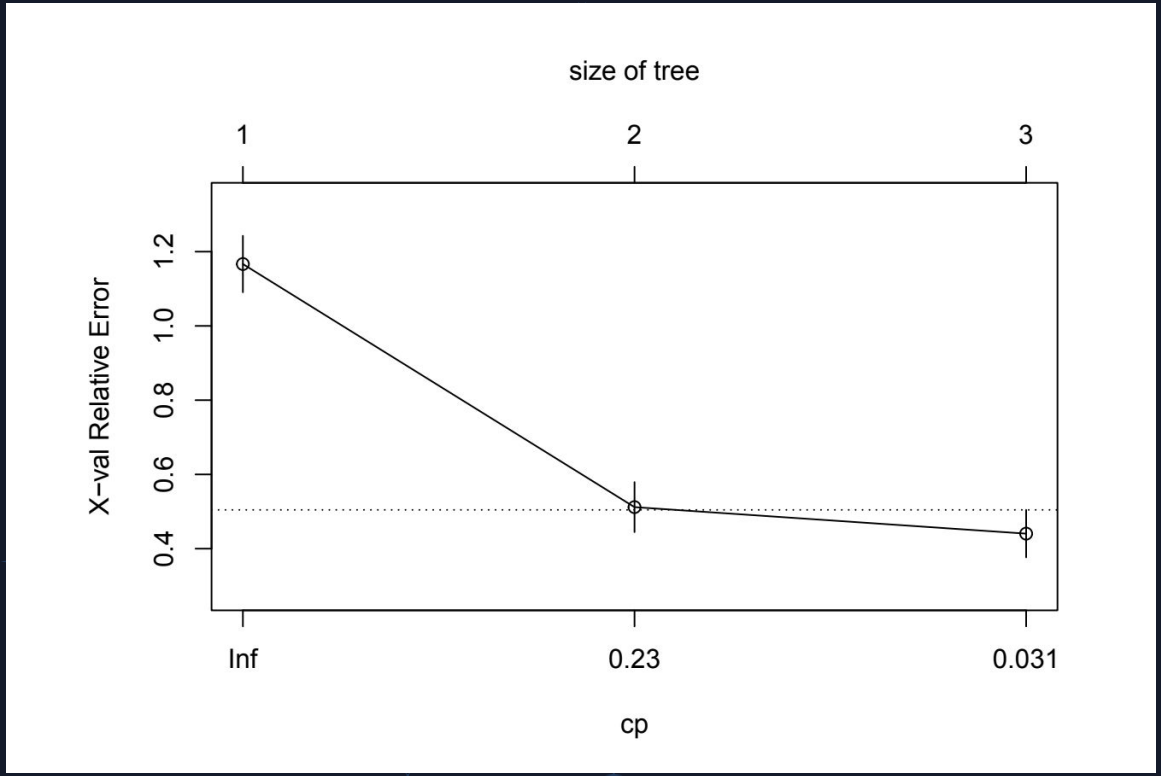
Single Classification Tree



CP plot

Best CP value: where cross-validation error (CV error) is lowest

In this case, best value is **around 0.031**, at **size 3**

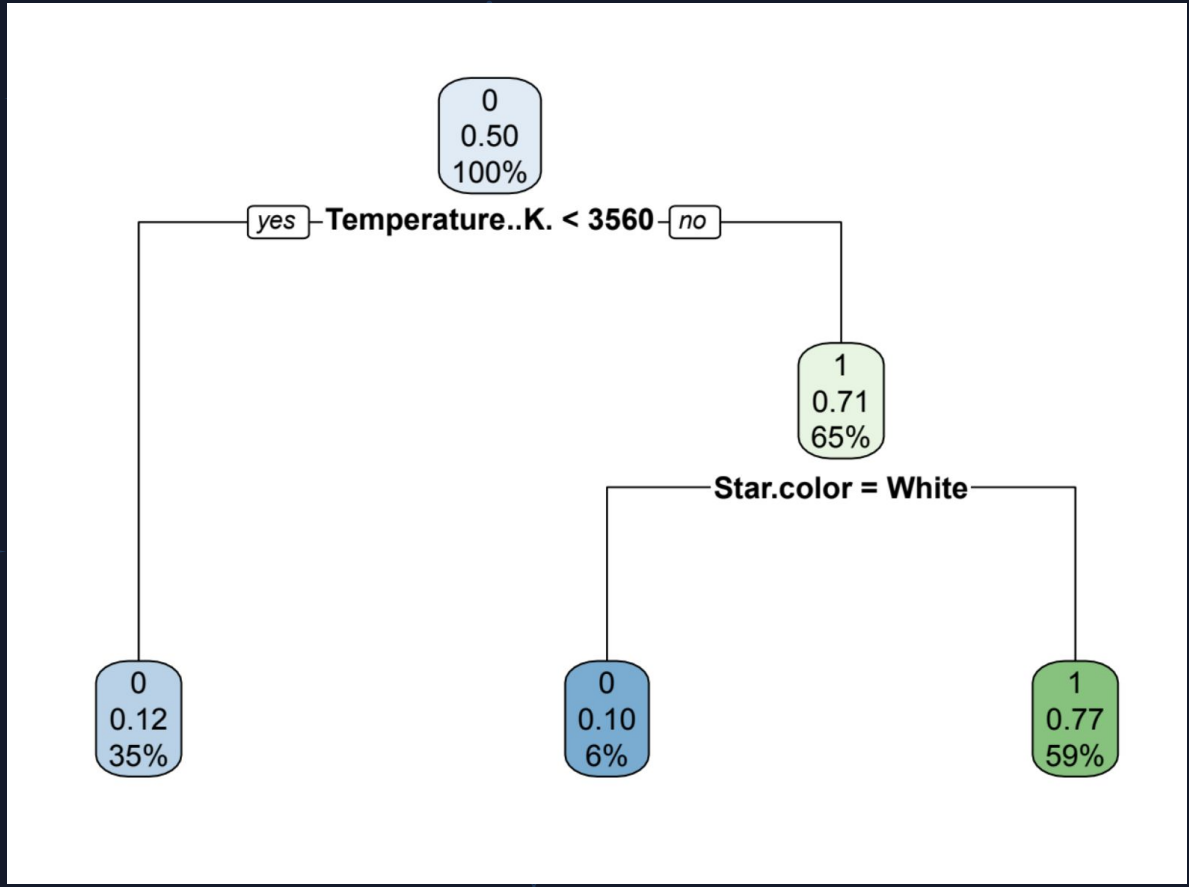


Prune the Tree

Pruned tree had **same structure** as the original

Confirms the original tree was **already optimal**

Model stayed simple and interpretable



✔️ 21 small stars correctly classified

✔️ 28 large stars correctly classified

❌ 15 large stars predicted as small

❌ 8 small stars predicted as large

Predictions made using both the full and pruned tree

Accuracy (both): 68.1%

Model performance stayed the same after pruning

```
[1] 0.6805556
```

```
##     predTree1
##        0   1
##    0 21  15
##    1   8 28
```

```
##     predTree2
##        0   1
##    0 21  15
##    1   8 28
```

04 Bagging Model



Bagging Model

Continuation of Tree Model

Bootstrap aggregation

- Random bootstrap samples (with replacement) of the data are used to create trees.
- By averaging a set of observations it reduces variance but you lose the interpretability

Predicting with Bagging Model

Accuracy: 83.3%

Using:

- Temperature
- Luminosity
- Star Color

	predBag					
	0	1	2	3	4	5
0	9	3	0	0	0	0
1	2	10	0	0	0	0
2	0	0	12	0	0	0
3	0	0	0	10	2	0
4	0	0	0	0	10	2
5	0	0	0	0	3	9



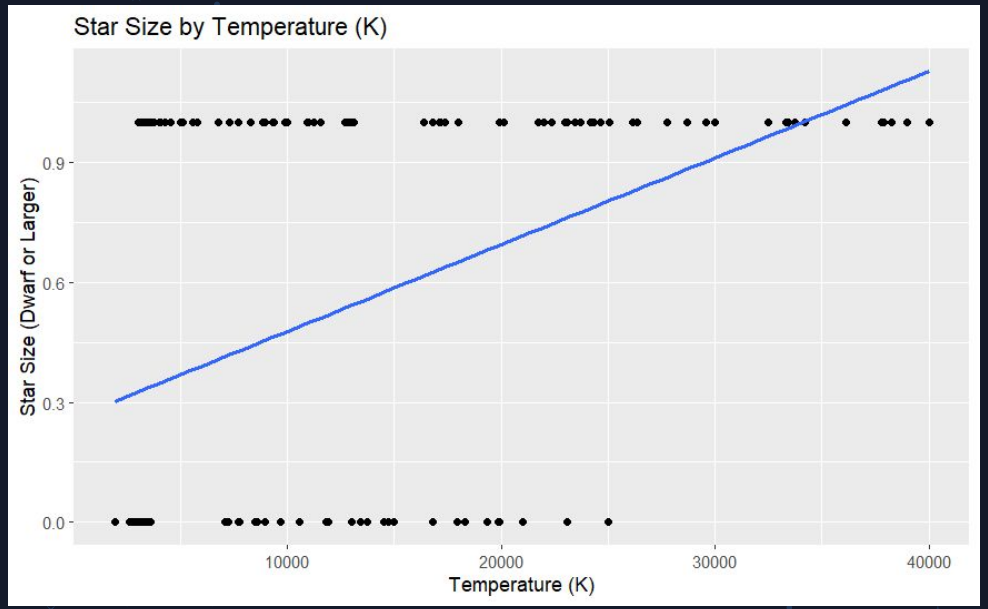
05

Logistic Regression

Logistic Regression Model

Needed 2 option categorical

Predicting: Star Size (Dwarf vs main sequence and Giants)

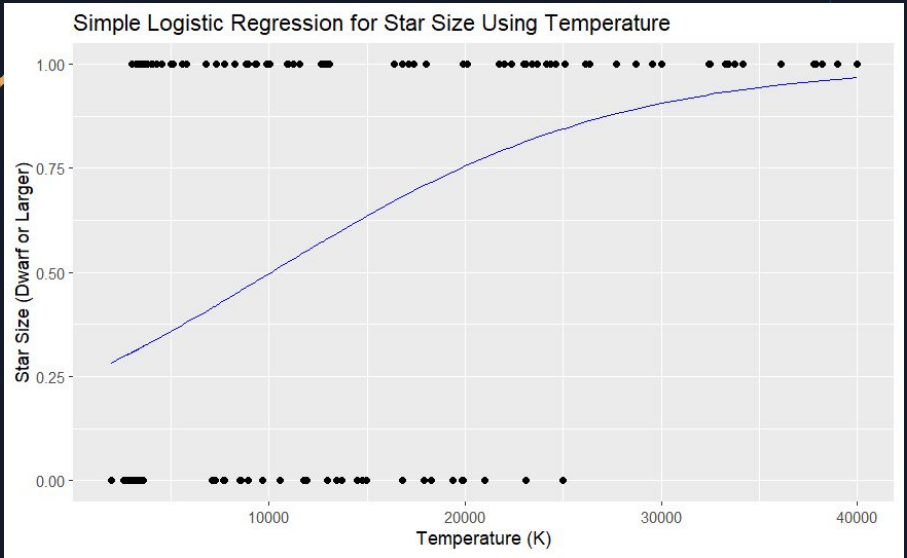


Why we can't use linear regression

Simple Linear Regression

Accuracy: 54.2%
Predicting with Temperature

	0	1
FALSE	25	22
TRUE	11	14




```
Call:
glm(formula = starSize ~ Temperature..K., family = "binomial",
    data = trainCaret)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    -1.151e+00  2.635e-01  -4.369 1.25e-05 ***
Temperature..K.  1.138e-04  2.263e-05   5.030 4.92e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 232.90  on 167  degrees of freedom
Residual deviance: 195.97  on 166  degrees of freedom
AIC: 199.97

Number of Fisher Scoring iterations: 4
```

Variable Selection

Backwards

```
Start:  AIC=16.01
planetSize ~ `Temperature (K)` + `Luminosity(L/Lo)` + Star.color

                     Df Deviance     AIC
- Star.color          5    3.615   9.615
<none>                     0.011  16.011
- `Temperature (K)`   1    2.062  16.062
- `Luminosity(L/Lo)`  1  179.673 193.673

Step:  AIC=9.61
planetSize ~ `Temperature (K)` + `Luminosity(L/Lo)`

                     Df Deviance     AIC
- `Temperature (K)`   1    3.691   7.691
<none>                     3.615   9.615
- `Luminosity(L/Lo)`  1  195.968 199.968

Step:  AIC=7.69
planetSize ~ `Luminosity(L/Lo)`

                     Df Deviance     AIC
<none>                     3.691   7.691
- `Luminosity(L/Lo)`  1  232.897 234.897

Call:  glm(formula = planetSize ~ `Luminosity(L/Lo)`, family = "binomial",
    data = select(trainCaret, -`Absolute magnitude(Mv)`, -`Radius(R/Ro)`,
        -`Star color`))

Coefficients:
       (Intercept)  `Luminosity(L/Lo)`
             -5.05               51.48

Degrees of Freedom: 167 Total (i.e. Null);  166 Residual
Null Deviance:      232.9
Residual Deviance: 3.691        AIC: 7.691
```

Forwards

```
Start:  AIC=234.9
planetSize ~ 1

                     Df Deviance     AIC
+ `Luminosity(L/Lo)`  1    3.691   7.691
+ Star.color          5  184.500 196.500
+ `Temperature (K)`   1  195.968 199.968
<none>                   232.897 234.897

Step:  AIC=7.69
planetSize ~ `Luminosity(L/Lo)`

                     Df Deviance     AIC
<none>                     3.691   7.691
+ `Temperature (K)`   1    3.615   9.615
+ Star.color          5    2.062  16.062
- `Luminosity(L/Lo)`  1  232.897 234.897

Call:  glm(formula = planetSize ~ `Luminosity(L/Lo)`, family = binomial,
    data = select(trainCaret, -`Absolute magnitude(Mv)`, -`Radius(R/Ro)`,
        -`Star color`))

Coefficients:
       (Intercept)  `Luminosity(L/Lo)`
             -5.05               51.48

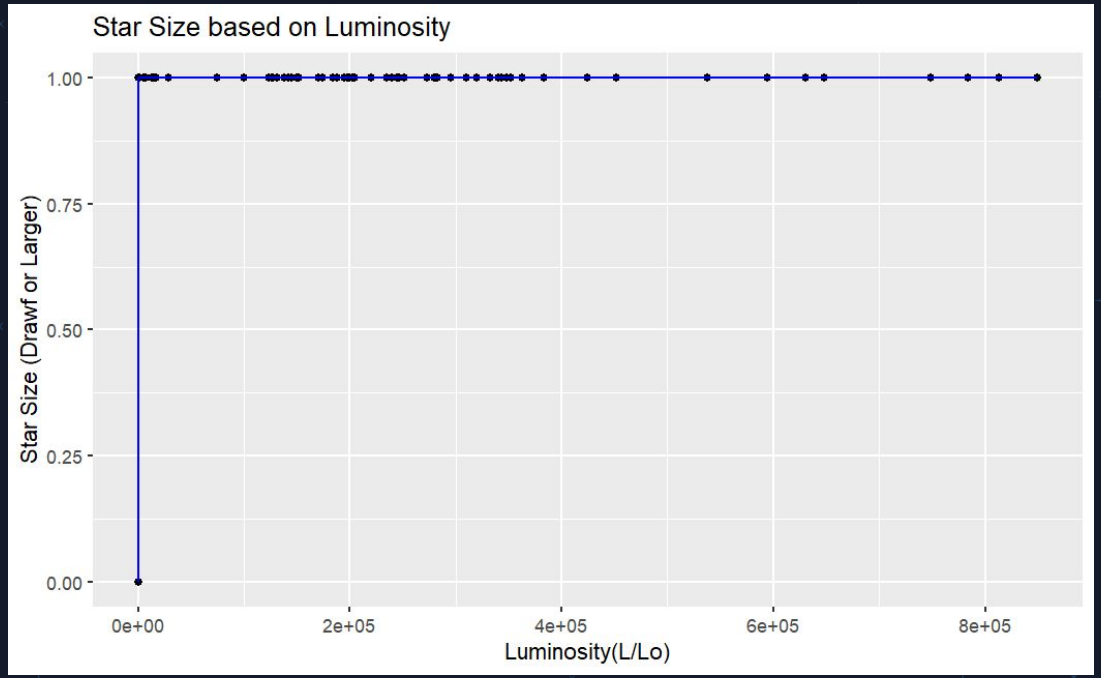
Degrees of Freedom: 167 Total (i.e. Null);  166 Residual
Null Deviance:      232.9
Residual Deviance: 3.691        AIC: 7.691
```

Best Logistic Model

Predicting Star Size based on Luminosity

Accuracy: 100%

	0	1
FALSE	36	0
TRUE	0	36

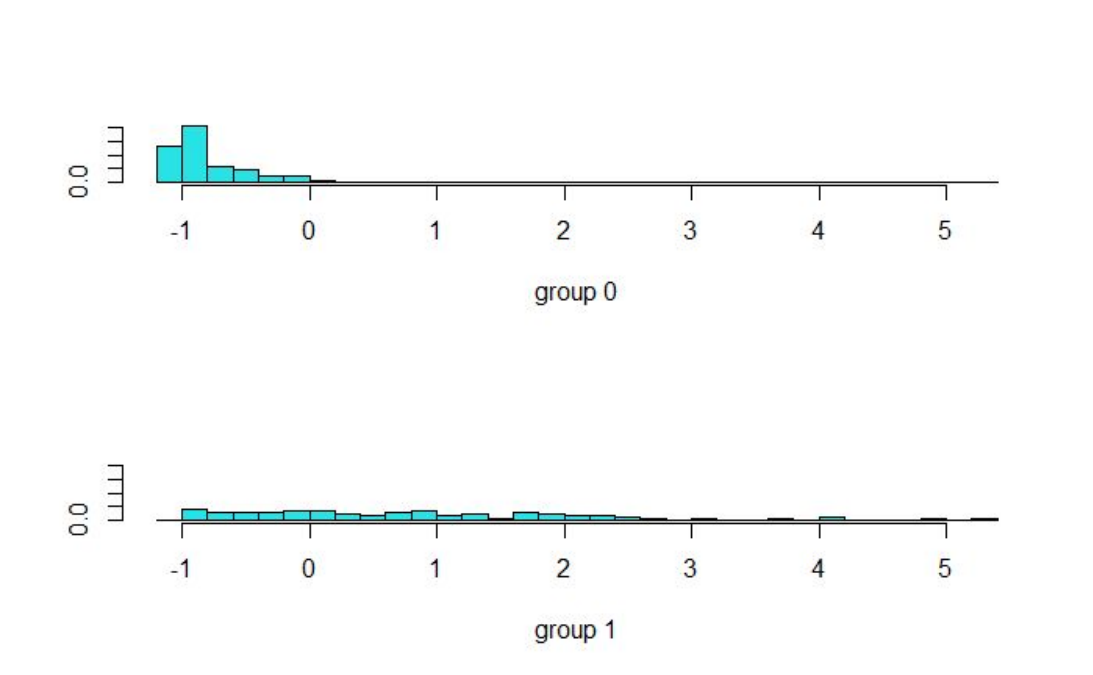


06

LDA and QDA Model



LDA Model Overview

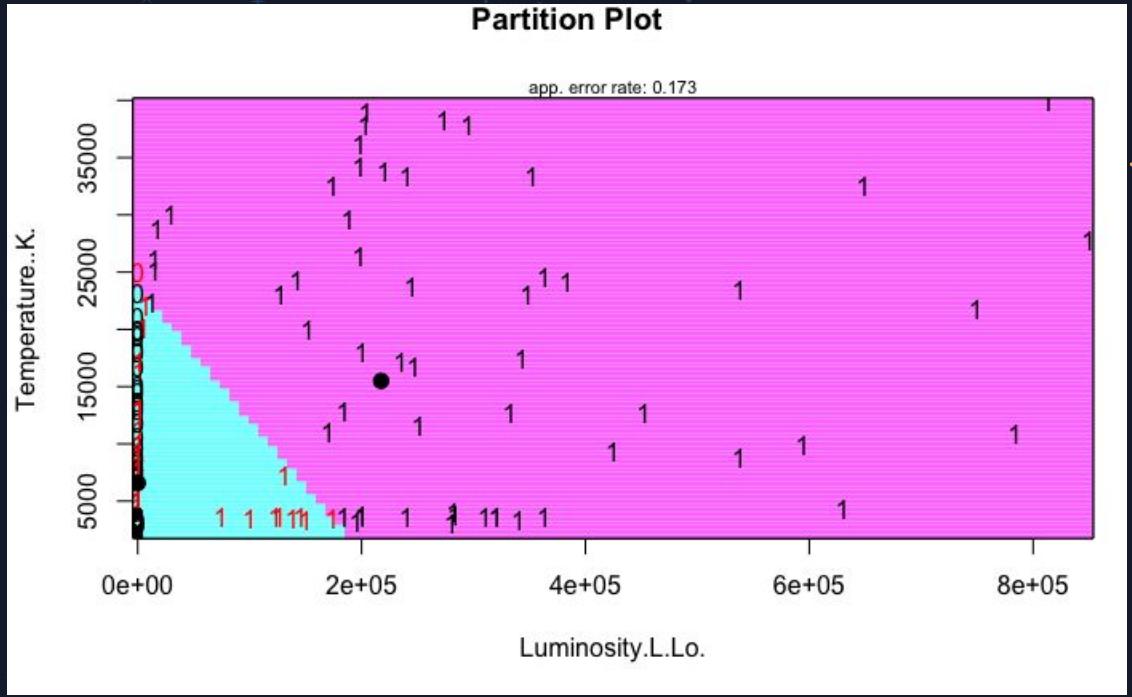


LDA Plot

Correct Rate

```
mean(predLDA$class==da.testCaret$starSize)
```

```
## [1] 0.827381
```

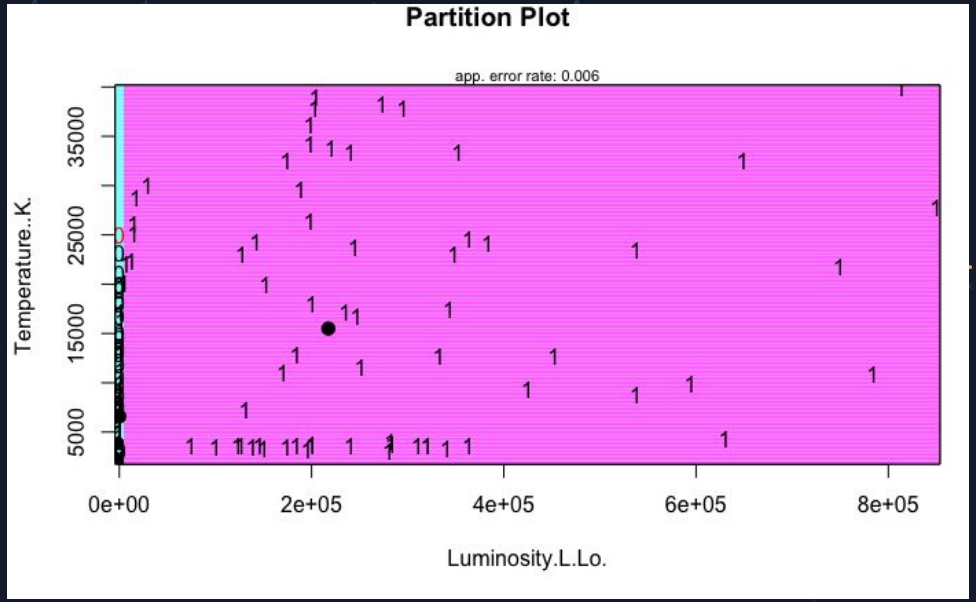


QDA Plot

Correct Rate

```
mean(predQDA$class==da.testCaret$starSize)
```

```
## [1] 0.9940476
```



07

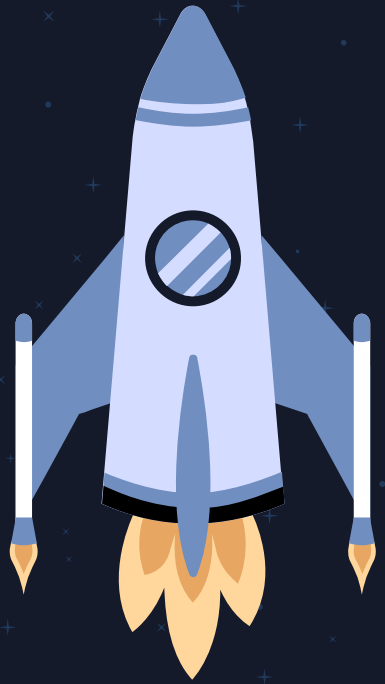
Conclusions

Conclusions

01 Star Type Predictor

KNN: 69%

Bagging Model: 83%



Binary Predictor 02

Tree Model: 68%

Logistic Regression: 100%*

QDA: 99%

LDA: 83%